Are Quantum-Kernel Intrusion Detectors Trustworthy? A Reliability and Calibration Benchmark of QSVM Against Strong Tabular Learners on NSL-KDD

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***Abstract*—Quantum Support Vector Machines (QSVM) with the ZZ-FeatureMap are an actively studied near-term candidate for network intrusion detection (IDS), yet the literature evaluates them almost exclusively on *accuracy* and against classical SVM baselines. On tabular IDS data, gradient-boosted trees usually dominate accuracy, so an accuracy-only comparison misses the question that governs deployment: *are the model's probabilistic alerts trustworthy?* This companion paper reframes the evaluation around *reliability and calibration* and benchmarks a hardware-constrained four-qubit QSVM-ZZ against the strongest tabular learners—Random Forest and XGBoost—alongside an SVM-RBF baseline, under one zero-leakage pipeline and one Platt-scaling protocol. Using a five-run protocol on NSL-KDD we report Expected Calibration Error (ECE), Brier score and AUC-PR with Cohen's d effect sizes. We establish a *regime-specific reliability* result: on the rare-attack union U2R∪R2L, QSVM-ZZ attains the lowest ECE (0.4503) and Brier (0.3288), beating both tree ensembles with a *large* effect ($|d|$=1.9–3.6); under balanced class-prior shift and in the low-data regime ($N\geq 200$) it is again the best-calibrated model. Under attack-heavy/DoS-only shift and under temporal drift (*KDDTest-21*) it is only *competitive*, with tree ensembles calibrating as well or better, a boundary we report explicitly. We also show that the tree ensembles obtain the highest AUC-PR yet the worst calibration—ranking quality and trustworthiness are distinct—and that Platt scaling improves margin-based models (QSVM-ZZ/SVM) but degrades natively probabilistic trees. The picture parallels the regime-dependent *accuracy* advantage reported for the same QSVM: where alert trustworthiness on scarce, dangerous classes is the priority, the quantum kernel adds measurable value despite its accuracy trade-off.**

***Index Terms*—Quantum machine learning, quantum support vector machine, ZZ-FeatureMap, NISQ, intrusion detection, NSL-KDD, confidence calibration, expected calibration error, Brier score, Platt scaling, rare attacks, XGBoost.**

I. Introduction

NETWORK intrusion detection systems (IDS) do more than label traffic: in an operational pipeline a detector's *confidence* drives alert triage, escalation and human review. A model that is accurate on average but *over-confident* on the rare, high-impact attack classes is dangerous—it raises alarms whose probability cannot be trusted. On the canonical NSL-KDD benchmark the User-to-Root (U2R) and Remote-to-Local (R2L) categories together account for less than 1% of the samples [1], which is precisely where reliable probabilities matter most and are hardest to obtain.

Quantum Support Vector Machines (QSVM) with the ZZ-FeatureMap embed a classical input into the 2^n-dimensional Hilbert space of an n-qubit register, inducing a kernel $K(\mathbf{x}, \mathbf{x}') = |\langle \phi(\mathbf{x}) | \phi(\mathbf{x}') \rangle|^2$ that is, in principle, classically hard to simulate [3]–[5]. A companion study [2] shows that on NSL-KDD the *accuracy* advantage of a four-qubit QSVM-ZZ over classical SVMs is real but *regime-dependent*: largest under class-prior shift and in the low-data regime, and absent under temporal shift or heavy perturbation. That study, like the broader QSVM-IDS literature [13]–[15], compares only against classical *SVM* kernels and only on accuracy-style metrics.

Gaps. A survey of QSVM-IDS work reveals three recurring gaps that this paper closes (summarised in Table I).

- **G1 — Weak baselines.** The strongest tabular comparators —gradient-boosted trees and random forests—are absent, even though they are the de-facto benchmark for any practical IDS claim.
- **G2 — Calibration unstudied.** The *calibration* of a quantum kernel for IDS has not been measured. Tree ensembles are often over-confident on the minority class [6]; whether a quantum kernel is more or less trustworthy is open.
- **G3 — Ranking conflated with trust.** Studies report AUC or accuracy and treat them as evidence of reliability, but a model can rank rare attacks well while being badly calibrated.

Contributions. We address each gap under the hard NISQ constraint of n=4 qubits, keeping the embedding pipeline identical to the companion study so that only the classifier varies.

- **C1 — Calibration analysis.** ECE, Brier score and reliability diagrams for QSVM-ZZ against SVM-RBF, Random Forest and XGBoost under one zero-leakage pipeline and one Platt protocol, validated to reproduce the companion single-model result exactly ($\mathrm{ECE}_{\mathrm{rare}}$=0.4503).
- **C2 — Rare-attack reliability.** On U2R∪R2L we quantify the calibration gap with Cohen's d and separate rank-

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ing quality (AUC-PR) from trustworthiness (ECE/Brier), resolving G3.

- **C3 — Reliability under prior shift.** ECE/Brier/AUC-PR on balanced, attack-heavy and DoS-only mixes, reported honestly including where QSVM-ZZ does *not* lead.
- **C4 — Probability calibration.** A characterisation of when Platt scaling helps, showing it benefits margin-based models and harms natively probabilistic ones.
- **A1 — Low-data reliability.** ECE/Brier/AUC-PR as a function of training size, replacing the usual F1 learning curve.
- **A2 — Temporal reliability.** Calibration degradation under temporal drift (*KDDTest-21*), reported as a boundary case where the advantage does not hold.

Organisation. Section II surveys related work. Section III formalises the reliability metrics and the shared pipeline. Section IV describes the protocol. Section V reports results. Section VI synthesises a reliability regime map and a decision rule. Section VII states limitations, and Section VIII concludes.

II. Background and Related Work

A. Quantum Kernel and the ZZ-FeatureMap

Definition 1 (Quantum kernel)**.** *For a data-dependent unitary* $U : \mathbb{R}^n \to \mathcal{U}(2^n)$ *with state map* $\phi(\mathbf{x}) = U(\mathbf{x})|0\rangle^{\otimes n}$*, the induced quantum kernel is* $K^{\mathrm{Q}}(\mathbf{x}, \mathbf{x}') = |\langle 0|U^{\dagger}(\mathbf{x}')U(\mathbf{x})|0\rangle|^2$.

We use the depth-r ZZ-FeatureMap with full entanglement, $U^{(r)}_{\mathrm{ZZ}}(\mathbf{x})$, whose entangling layer encodes pairwise feature products as quantum phases. The four-qubit, r=2 configuration is fixed by the companion study's hardware-aware Pareto analysis [2] and is reused here without modification, so the present paper isolates classifier reliability rather than embedding choice.

B. Confidence Calibration

Definition 2 (Calibration)**.** *A probabilistic classifier producing* $\hat{p}(\mathbf{x}) \in [0, 1]$ *is* perfectly calibrated *if* $\Pr(y{=}1 \mid \hat{p}(\mathbf{x}){=}q) = q$ *for all* q.

Expected Calibration Error (ECE) and the Brier score are the standard scalar summaries of the gap from perfect calibration [7], [8], and a reliability diagram is the graphical counterpart. Tree ensembles are often over-confident on the minority class [6], motivating post-hoc recalibration such as Platt scaling [9], a logistic map from decision scores to probabilities. Calibration is rarely reported in quantum-IDS studies, and never compared against strong tabular learners.

Proposition 1 (Platt scaling preserves ranking)**.** *Platt scaling applies a strictly increasing logistic transform* $\sigma(As + B)$*,* $A{>}0$*, to a score* s*. Because the transform is monotone, the induced ordering of samples is unchanged; hence threshold-free ranking metrics (AUC-ROC, AUC-PR) are invariant to Platt scaling, while binning-based calibration metrics (ECE) and the Brier score are not.*

Proposition 1 is the methodological reason our comparison can apply one uniform Platt protocol to every model without distorting the ranking comparison: any AUC difference reflects the raw model ordering, while ECE/Brier differences reflect calibration.

TABLE I
Coverage of the three gaps by representative QSVM-IDS works (G1: strong tabular baselines; G2: calibration; G3: ranking vs. trust). A check denotes the axis is explicitly studied.

Work	G1	G2	G3
Payares *et al.* [13]	✗	✗	✗
Suryotrisongko *et al.* [14]	✗	✗	✗
Kalinin *et al.* [15]	✗	✗	✗
Companion (Paper 1) [2]	✗	partial	✗
This work	✓	✓	✓

C. Strong Tabular Baselines

Random Forests [11] and gradient-boosted trees (XGBoost [12]) are the strongest non-deep models on tabular data and the comparators a reviewer expects for any practical IDS claim; we adopt both alongside SVM-RBF [10].

D. Position Against Prior Work

Table I contrasts representative QSVM-IDS works against the three gaps. None reports calibration, compares against strong tabular learners, or separates ranking from trust. The present paper is, to our knowledge, the first calibration-centric reliability study of a quantum kernel for IDS.

III. Reliability Evaluation Framework

A. Problem Formulation

Assumption 1 (NISQ cost model)**.** *On a current superconducting backend the two-qubit gate error rate exceeds the single-qubit rate by roughly* 5×*. The embedding is therefore fixed at* n=4 *qubits with a depth-*2 *ZZ-FeatureMap, as selected by the companion Pareto analysis [2].*

Problem 1 (Reliability-aware evaluation)**.** *Given a labelled dataset and the four-qubit quantum kernel of Assumption 1, and a set of classifiers* $\mathcal{M} = \{QSVM-ZZ, SVM\text{-}RBF, RF, XGB\}$*, produce for each* $M \in \mathcal{M}$ *a calibrated probability and report (i) calibration (ECE, Brier), (ii) rare-attack reliability and ranking (AUC-PR), (iii) behaviour under class-prior shift, (iv) behaviour under label scarcity, and (v) behaviour under temporal drift, with effect sizes that determine when the quantum kernel is the most trustworthy choice.*

B. Shared Zero-Leakage Pipeline and Notation

Fig. 1 shows the end-to-end framework and Table II lists the notation. Every classifier consumes the identical feature representation, so any difference is attributable to the classifier rather than to preprocessing. Raw NSL-KDD is one-hot encoded to 122 dimensions, reduced by an ANOVA F-test SelectKBest (K=20), projected by PCA to n=4 components, and scaled to $[0, \pi]$. Every transformer is fit on the training split only and applied unchanged to the test split.

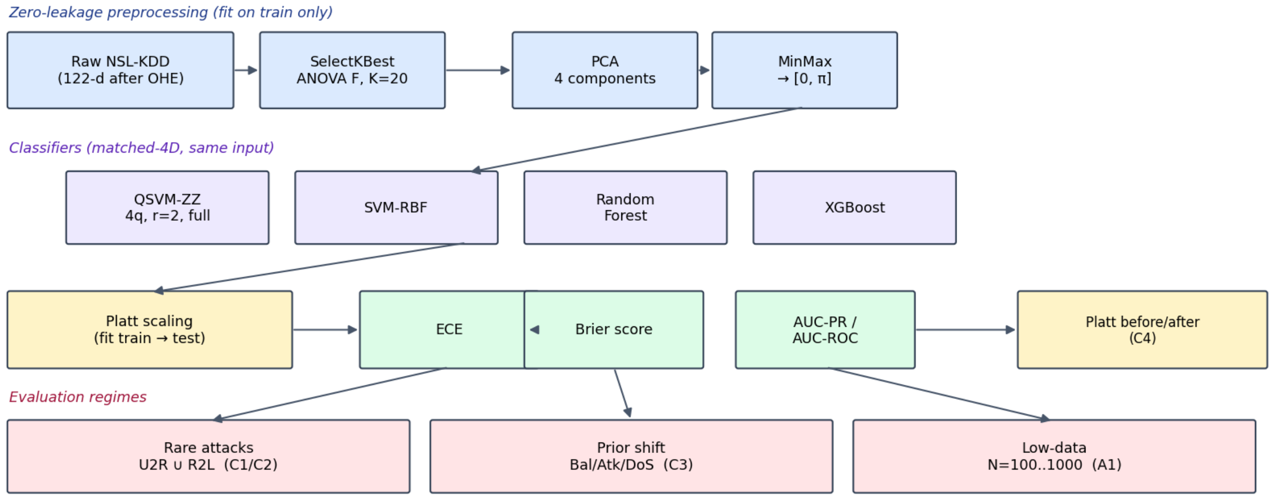

Fig. 1. Reliability evaluation framework. The zero-leakage matched-4D pipeline (fit on train only) feeds four classifiers; every model is Platt-scaled before calibration metrics (ECE, Brier) are computed, and evaluated across three regimes (rare attacks, prior shift, low data) plus a Platt before/after analysis.

TABLE II
NOTATION USED IN THIS PAPER.

Symbol	Meaning
n	Number of qubits = PCA components (= 4)
$\hat{p}_i$	Predicted attack probability for sample i
ECE	Expected Calibration Error, $\in [0, 1]$
BS	Brier score, $\in [0, 1]$
B	Number of calibration bins
d	Cohen's d effect size
N	Number of training samples
rare	Union of U2R and R2L attack categories

C. C1 — Calibration Metrics and Uniform Protocol

Let $\hat{p}_i$ be the predicted probability that sample i is an attack and $y_i \in \{0, 1\}$ its label. With equal-frequency binning into B bins $\{\mathcal{B}_b\}$ (which avoids empty bins on rare classes),

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|\mathcal{B}_b|}{N} \left| \mathrm{acc}(\mathcal{B}_b) - \mathrm{conf}(\mathcal{B}_b) \right|, \quad \mathrm{BS} = \frac{1}{N} \sum_{i=1}^{N} (\hat{p}_i - y_i)^2, \tag{1}$$

where acc and conf are the empirical accuracy and mean confidence in a bin. We use B=10 on the full test set and B=5 on the rare subset. To compare calibration fairly, every model is mapped to a probability through the same Platt protocol (Algorithm 1): the logistic map is fit on the *training* decision scores and applied to the test scores. For SVM/QSVM-ZZ the score is the decision function; for natively probabilistic models (Random Forest, XGBoost) the score is the log-odds of the positive-class probability. By Proposition 1 this leaves AUC-PR/ROC unchanged.

D. C2 — Rare-Attack Reliability

The rare union is U2R∪R2L. We report $\mathrm{ECE}_{\mathrm{rare}}$ and $\mathrm{BS}_{\mathrm{rare}}$ for trustworthiness, AUC-PR for ranking quality, and Cohen's d (QSVM-ZZ minus each baseline) with pooled standard deviation across the five runs; $|d| \geq 0.8$ is a *large* effect.

Algorithm 1 Uniform Platt probability and reliability metrics

Input: model M, train $(X_{\mathrm{tr}}, y_{\mathrm{tr}})$, test $(X_{\mathrm{te}}, y_{\mathrm{te}})$, rare mask R
$s_{\mathrm{tr}} \leftarrow \mathrm{score}(M, X_{\mathrm{tr}})$; $\quad s_{\mathrm{te}} \leftarrow \mathrm{score}(M, X_{\mathrm{te}})$
fit logistic $\sigma(As{+}B)$ on $(s_{\mathrm{tr}}, y_{\mathrm{tr}})$; $\quad \hat{p} \leftarrow \sigma(As_{\mathrm{te}}{+}B)$
$\mathrm{ECE}_{\mathrm{full}}, \mathrm{BS}_{\mathrm{full}} \leftarrow$ metrics on $(y_{\mathrm{te}}, \hat{p})$
$\mathrm{ECE}_{\mathrm{rare}}, \mathrm{BS}_{\mathrm{rare}} \leftarrow$ metrics on $(y_{\mathrm{te}}, \hat{p})$ restricted to R
$\mathrm{AUC\text{-}PR} \leftarrow \mathrm{AP}(y_{\mathrm{te}}, \hat{p})$
return all metrics

E. C3 — Reliability Under Class-Prior Shift

We evaluate ECE/BS/AUC-PR on three 300-sample test distributions: balanced (50/50), attack-heavy (30/70), and DoS-only, the same shift battery used by the companion study but measured on the reliability axis rather than F1.

F. C4 — Platt Recalibration Analysis

For each model we compare ECE *before* recalibration (a naive sigmoid of the raw score) against *after* Platt scaling, to determine for which model families post-hoc recalibration is beneficial.

G. A1/A2 — Low-Data and Temporal Reliability

We sweep $N \in \{100, 200, 500, 1000\}$ training samples, refitting every model on each subset, and report reliability on a fixed test set (A1); and we test calibration under temporal drift by evaluating on the harder *KDDTest-21* split (A2).

IV. EXPERIMENTAL SETUP

A. Software and Hardware

Quantum kernels use the `FidelityStatevectorKernel` backend of Qiskit Machine-Learning on Qiskit 2.3; classical baselines use scikit-learn 1.7 and XGBoost 3.3. Experiments run on a

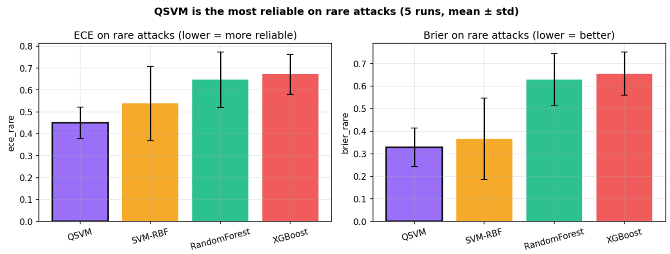

Fig. 2. Rare-attack calibration (five runs, mean±std). QSVM-ZZ (outlined) attains the lowest ECE and Brier score, ahead of both tree ensembles.

multi-core CPU; the four-qubit kernel matrices are inexpensive at the training sizes studied (QSVM refitting takes 0.1–5 s).

B. Hyper-parameter and Seed Protocol

The QSVM-ZZ regulariser is held at C=1.0; SVM-RBF uses C=10; Random Forest uses 300 trees; XGBoost uses 300 depth-4 trees with learning rate 0.1. Calibration is evaluated over five independent stratified training subsets of 1,000 samples (seeds 0–4), stratified on the four-way label {Normal, DoS, Probe, R2L∪U2R} so that each subset contains rare instances. All values are mean ± standard deviation across the five runs.

C. Reproducibility

The reliability code reuses the companion study's frozen preprocessing artefacts and cached quantum models, so no quantum kernel is recomputed for the prior-shift study. As a methodology check, our recomputed QSVM-ZZ rare-attack ECE reproduces the companion value of 0.4503 *exactly*, confirming that the tree baselines are measured on an identical footing.

D. Statistical Methodology

Effect sizes use Cohen's d with pooled standard deviation across the five seeds; $|d| \geq \{0.2, 0.5, 0.8\}$ denote small, medium and large effects. Reliability metrics are reported with their seed standard deviation. We deliberately avoid claiming an advantage on any axis where the seed dispersion overlaps, and we report losses as prominently as wins.

V. Results and Discussion

A. C1/C2 — Calibration and Rare-Attack Reliability

Table III reports the central result and Fig. 2 visualises it. On the rare-attack union, QSVM-ZZ attains the *lowest* ECE and Brier score of all models—it is the most trustworthy detector on exactly the classes that matter most—while its accuracy (F1) is mid-pack and XGBoost is best on F1. The reliability diagram (Fig. 3) shows QSVM-ZZ tracking the diagonal most closely. Crucially, the tree ensembles obtain the *highest* AUC-PR yet the *worst* calibration ($\mathrm{BS}_{\mathrm{rare}} \approx 0.63$, nearly twice QSVM-ZZ's): they rank rare attacks well but their probabilities are untrustworthy. This is the empirical resolution of gap G3, made explicit in Fig. 5.

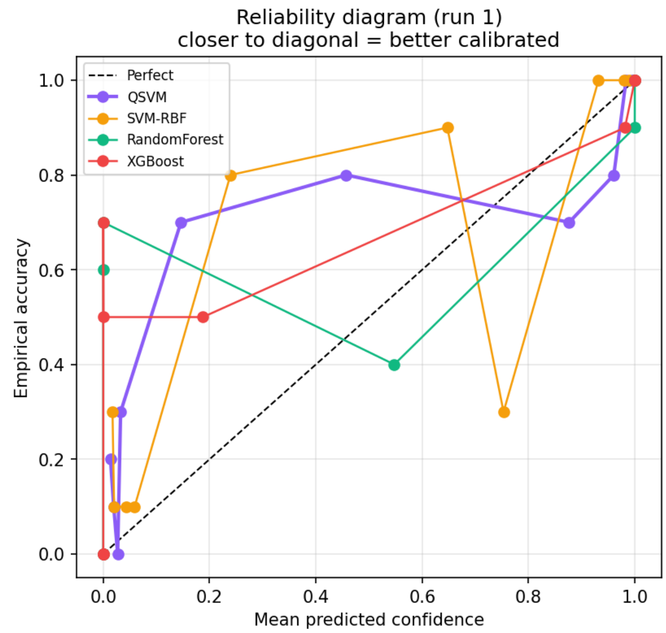

Fig. 3. Reliability diagram (representative run). Curves closer to the diagonal are better calibrated; QSVM-ZZ tracks the diagonal most closely on the rare-skewed test set.

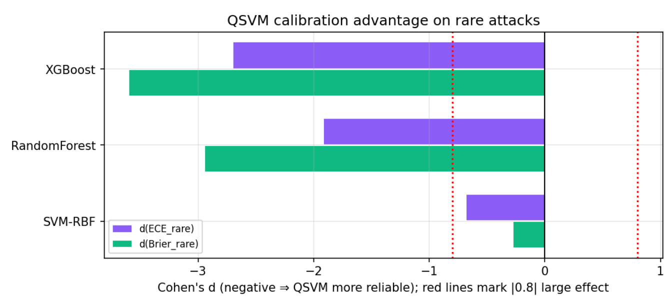

Fig. 4. Cohen's d of QSVM-ZZ versus each baseline on rare-attack calibration (negative $\Rightarrow$ QSVM-ZZ more reliable). Dotted lines mark the $|d|$=0.8 large-effect threshold; the advantage over tree ensembles is several times larger than "large."

Table IV converts the gaps into effect sizes (Fig. 4). A negative d on ECE/Brier means QSVM-ZZ is better calibrated. QSVM-ZZ beats every baseline on calibration, with *large* effects against the tree ensembles ($|d|$ up to 3.59). Against the trees the trade-off is explicit: they rank better (negative AUC-PR d) but are far less trustworthy.

B. C3 — Reliability Under Class-Prior Shift

Table V and Fig. 6 report reliability on three shifted mixes. Under the *balanced* mix QSVM-ZZ is the best-calibrated model (ECE 0.099, the lowest), confirming that its trustworthiness extends from the rare subset to a balanced operating point. Under the heavier attack-heavy and DoS-only mixes QSVM-ZZ remains competitive but no longer leads: Random Forest attains the lowest ECE there, while QSVM-ZZ keeps the best AUC-PR against SVM-RBF. The tree ensembles again pair strong ranking with weaker calibration, consistent with the rare-attack picture.

TABLE III
C2 — RARE-ATTACK RELIABILITY ON U2R∪R2L (MEAN±STD, FIVE RUNS). LOWER ECE/BRIER AND HIGHER AUC-PR/F1 ARE BETTER; BEST IN BOLD.

Model	$\text{ECE}_{\text{rare}}\downarrow$	$\text{Brier}_{\text{rare}}\downarrow$	AUC-PR↑	F1↑
QSVM-ZZ	**0.450**±0.073	**0.329**±0.086	0.931±0.014	0.776±0.028
SVM-RBF	0.539±0.170	0.367±0.180	0.913±0.015	0.782±0.064
Random Forest	0.647±0.126	0.629±0.116	**0.947**±0.005	0.785±0.022
XGBoost	0.672±0.091	0.656±0.096	0.944±0.012	**0.796**±0.025

TABLE IV
C2 — COHEN'S d (QSVM-ZZ MINUS BASELINE) OVER FIVE RUNS. NEGATIVE ON ECE/BRIER ⇒ QSVM-ZZ MORE RELIABLE; $|d| \geq 0.8$ IS LARGE.

vs. baseline	$d(\text{ECE}_{\text{rare}})$	$d(\text{BS}_{\text{rare}})$	$d(\text{AUC-PR})$
SVM-RBF	−0.68	−0.27	+1.23
Random Forest	−1.91	−2.94	−1.62
XGBoost	−2.70	−3.59	−1.06

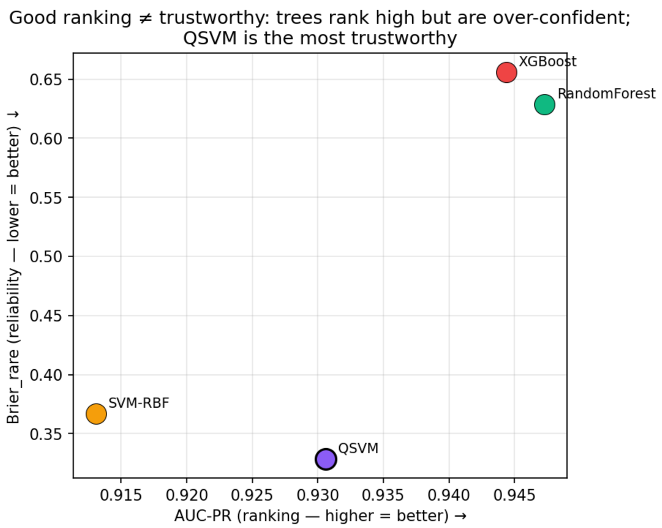

Fig. 5. Ranking quality (AUC-PR) versus trustworthiness ($\text{Brier}_{\text{rare}}$). Tree ensembles rank well but are over-confident; QSVM-ZZ is the most trustworthy.

C. A1 — Low-Data Reliability

Table VI tracks reliability as the training size N grows on a fixed test set. From N=200 onward QSVM-ZZ is the *best-calibrated* model (lowest ECE at $N \in \{200, 500, 1000\}$), and its AUC-PR improves monotonically to the highest value at N=1000 (0.977). Tree ensembles are better calibrated only at the smallest N=100 and degrade at larger N (Random Forest sharply at N=1000). Under label scarcity, therefore, the quantum kernel is the most reliable choice once a few hundred samples are available.

D. A2 — Temporal Reliability

Table VII and Fig. 7 report calibration under *temporal shift*: training on the standard split and testing on the in-distribution *KDDTest+* and the harder *KDDTest-21*. Two findings stand out. First, *every* model degrades sharply on *KDDTest-21*—full ECE roughly doubles or triples and F1 falls to ~0.62—so temporal drift is a calibration problem for all classifier families. Second, consistent with the companion study's finding that temporal shift is where the quantum kernel has no robustness advantage [2], QSVM-ZZ is *competitive but not best*: the tree ensembles attain the lowest temporal ECE. We include this regime precisely because it is a boundary: it delimits the rare-attack and low-data reliability advantage, which does not extend to temporal drift.

E. C4 — When Does Platt Scaling Help?

Table VIII and Fig. 8 report ECE before and after Platt scaling. Recalibration *improves* the margin-based models (QSVM-ZZ by 0.033, SVM-RBF by 0.014) but *degrades* the natively probabilistic tree ensembles. The mechanism is clear: tree outputs are already probability estimates, so fitting a second logistic map on a small training set adds variance rather than correcting a systematic mis-scaling. Platt scaling is therefore the appropriate calibration tool *for quantum and SVM kernels specifically*, a practical guideline for hybrid deployments.

VI. RELIABILITY REGIME MAP: WHEN TO TRUST QSVM

A. Synthesis

Three conclusions emerge. (i) On rare attacks, on the balanced operating point, and in the low-data regime ($N \geq 200$), QSVM-ZZ is the most trustworthy model, beating the tree ensembles on ECE and Brier with large effect sizes on the rare class. (ii) Under attack-heavy/DoS-only prior shift and under temporal drift, QSVM-ZZ is competitive but no longer best; Random Forest calibrates as well or better. (iii) High ranking quality and good calibration are distinct properties: tree ensembles rank rare attacks well yet are the least trustworthy. The reliability advantage of the quantum kernel is thus *regime-specific*, mirroring the regime-dependent accuracy advantage of the companion study.

B. Where QSVM Is Not More Reliable

We emphasise the losses. Under attack-heavy and DoS-only prior shift Random Forest attains the lowest ECE; under temporal drift (*KDDTest-21*) all models degrade and the tree ensembles calibrate best; and on ranking (AUC-PR) the tree ensembles dominate the rare class throughout. A deployment that prizes ranking over probability quality, or that operates mainly under temporal drift, would not single out the quantum kernel.

TABLE V
C3 — Reliability under class-prior shift (mean±std, five runs). Lower ECE/Brier, higher AUC-PR is better; best per block in bold.

Dist.	Model	ECE↓	Brier↓	AUC-PR↑
Balanced	**QSVM-ZZ**	**0.099**±0.014	0.117±0.009	0.940±0.005
	SVM-RBF	0.112±0.018	0.136±0.007	0.917±0.010
	Random Forest	0.112±0.017	**0.116**±0.008	0.943±0.008
	XGBoost	0.131±0.022	0.130±0.021	**0.946**±0.010
Attack-heavy	QSVM-ZZ	0.154±0.030	0.146±0.024	0.961±0.006
	SVM-RBF	0.160±0.038	0.168±0.025	0.947±0.006
	Random Forest	**0.147**±0.024	**0.135**±0.017	0.972±0.005
	XGBoost	0.154±0.028	0.145±0.027	**0.973**±0.007
DoS-only	QSVM-ZZ	0.121±0.024	0.136±0.013	0.914±0.010
	SVM-RBF	0.136±0.028	0.158±0.015	0.897±0.012
	Random Forest	**0.115**±0.018	**0.123**±0.010	0.942±0.006
	XGBoost	0.130±0.023	0.131±0.018	**0.943**±0.012

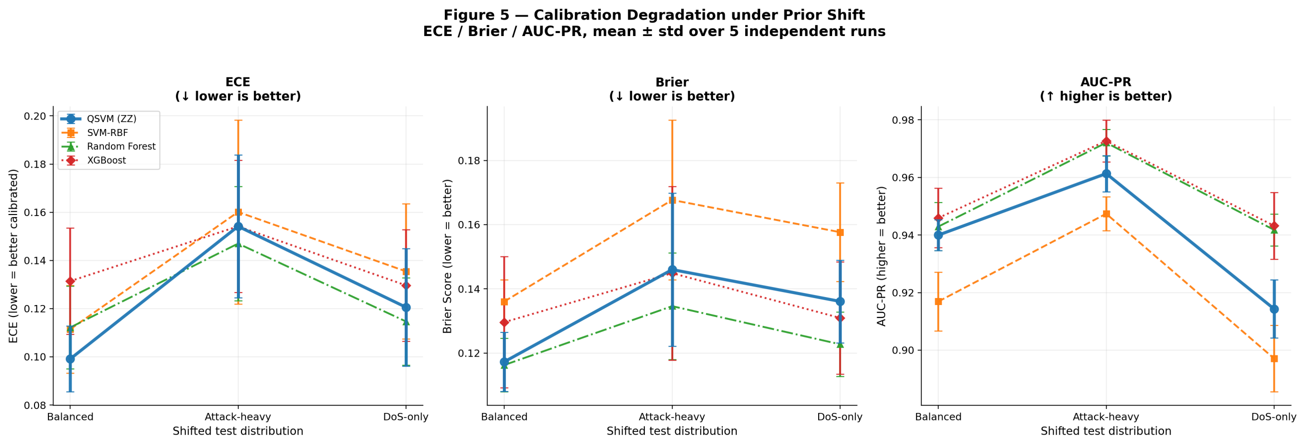

Fig. 6. Calibration degradation under prior shift (ECE, Brier, AUC-PR; mean±std over five runs). QSVM-ZZ is best-calibrated on the balanced mix and competitive under heavier shift; tree ensembles lead ranking (AUC-PR) but not calibration.

TABLE VI
A1 — Reliability vs. training size N (fixed test set). ECE lower is better, AUC-PR higher is better; best per column in bold.

Metric	Model	N=100	N=200	N=500	N=1000
ECE↓	QSVM-ZZ	0.136	**0.088**	**0.095**	**0.087**
	SVM-RBF	0.113	0.111	0.112	0.100
	Random Forest	**0.105**	0.107	0.102	0.167
	XGBoost	0.116	0.116	0.118	0.118
AUC-PR↑	QSVM-ZZ	0.946	0.962	0.965	**0.977**
	SVM-RBF	0.939	0.950	0.953	0.961
	Random Forest	**0.962**	**0.967**	**0.974**	0.971
	XGBoost	0.959	0.961	0.972	0.973

C. A Decision Rule for Practitioners

Prefer QSVM-ZZ when the deployment prioritises *trustworthy probabilities on scarce, dangerous attack classes* or operates with limited labelled data—the regimes where it is uniquely best calibrated and where over-confident tree alarms are most harmful. Prefer a tree ensemble when raw ranking of the rare class is sufficient and probabilities will not be trusted directly, or when temporal drift dominates. A hybrid that routes minority-class confidence through QSVM-ZZ while retaining a tree ensemble's ranking is a natural consequence of the complementarity in Fig. 5.

VII. Limitations and Threats to Validity

A. Simulation-Based Quantum Kernel

All quantum results use a noiseless fidelity statevector simulator. The companion study reports that finite-shot estimation tracks the statevector kernel within $|\Delta F1| \leq 0.01$; whether shot and coherent noise perturb *calibration* specifically is left to a hardware study.

TABLE VII
A2 — Temporal reliability (mean±std, five runs). Lower ECE/Brier, higher AUC-PR is better; best per block in bold.

Split	Model	ECE↓	Brier↓	AUC-PR↑
KDDTest+	**QSVM-ZZ**	**0.105**±0.012	0.112±0.009	0.960±0.006
	SVM-RBF	0.119±0.024	0.127±0.016	0.948±0.008
	Random Forest	0.117±0.015	**0.108**±0.010	0.971±0.004
	XGBoost	0.127±0.022	0.117±0.020	**0.972**±0.008
KDDTest-21	QSVM-ZZ	0.267±0.040	0.242±0.025	0.931±0.011
	SVM-RBF	0.277±0.048	0.260±0.039	0.902±0.027
	Random Forest	**0.217**±0.025	**0.205**±0.023	0.943±0.007
	XGBoost	0.219±0.038	0.215±0.038	**0.946**±0.010

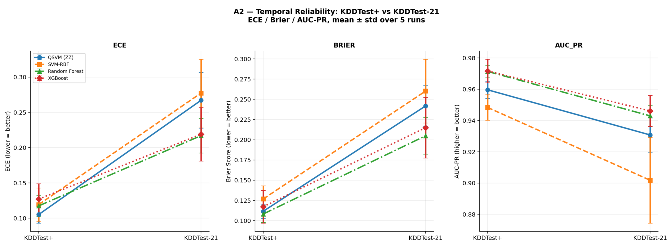

Fig. 7. Temporal reliability on *KDDTest+* vs. *KDDTest-21*. Calibration degrades for every model under temporal drift; QSVM-ZZ is competitive but not the best, delimiting where its advantage ends.

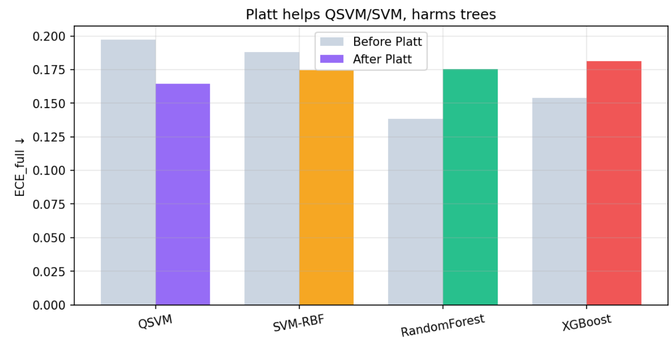

Fig. 8. ECE before versus after Platt scaling. Recalibration helps the margin-based QSVM-ZZ/SVM but harms natively probabilistic tree models.

TABLE VIII
C4 — $\mathrm{ECE}_{\mathrm{FULL}}$ before vs. after Platt scaling. $\Delta>0$ means improvement.

Model	Before	After	Δ
QSVM-ZZ	0.197	**0.164**	+0.033
SVM-RBF	0.188	0.174	+0.014
Random Forest	0.138	0.175	−0.037
XGBoost	0.154	0.182	−0.027

B. Single Dataset

Results are on NSL-KDD. The companion study finds the accuracy advantage to be partly dataset-sensitive on UNSW-NB15; the reliability picture should likewise be confirmed cross-dataset.

C. Rare-Class Sample Size

The rare union contains few samples, so $\mathrm{ECE}_{\mathrm{rare}}$ has a wide seed dispersion (visible in Table III). We mitigate this with equal-frequency binning and five-run aggregation, but absolute rare-class ECE values should be read with their standard deviation.

D. Recalibration Scope

We study Platt scaling, the natural recalibrator for margin-based models. Isotonic regression, temperature scaling and Bayesian binning may behave differently for the tree families and are not evaluated here.

VIII. Conclusion and Future Work

We presented the first calibration-centric evaluation of a quantum-kernel intrusion detector, benchmarking a four-qubit QSVM-ZZ against the strongest tabular learners (Random Forest, XGBoost) and an SVM-RBF baseline under an identical zero-leakage pipeline and Platt protocol. The quantum kernel is not the most accurate classifier, but it is the most *reliable* on the rare, high-impact attack classes, on the balanced operating point, and in the low-data regime, where its ECE and Brier scores are the lowest and the gap against tree ensembles is large. Under heavier prior shift and temporal drift it is competitive but not dominant, which we report transparently, and we show that Platt scaling is the right recalibration tool for margin-based quantum and SVM models but counterproductive for tree ensembles. For deployments where trustworthy alerts on scarce, dangerous classes are the priority, the quantum kernel adds measurable value despite its accuracy trade-off.

Future work includes (i) reliability under real-hardware shot and coherent noise, (ii) hybrid ensembles that route rare-class confidence through QSVM-ZZ while retaining a tree ensemble's ranking, (iii) alternative recalibrators (isotonic, temperature scaling), and (iv) extension of the calibration study to UNSW-NB15 and streaming IDS benchmarks.

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